

Andres Mercado

BACKEND SOFTWARE ENGINEER

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Backend Software Engineer specializing in async Python systems, job-based architectures, and data-intensive APIs. Experience building stateless integrations at a YC-backed startup using Flask, Google Cloud Run, and webhook-driven workflows. Designed multi-service FastAPI platforms leveraging PostgreSQL, vector embeddings, and GPU-backed OCR pipelines, with a strong focus on system design and concurrency control.

LANGUAGES: Python, JavaScript, SQL

FRAMEWORKS: Flask (with WTForms), Express, FastAPI

Cloud & DevOps: AWS (EC2, S3), Google Cloud Run, Docker, Nginx

SPOKEN LANGUAGES: English, Spanish (Fluent)

DATABASES: PostgreSQL, SQLite, MongoDB

TOOLS: GitHub, Postman, SQLAlchemy

WORK EXPERIENCE

Stacksync >> Backend Engineer (Contract) > Remote

June 2025 – September 2025

Python, Flask, Google Cloud Run, Webhooks

- **Delivered stateless API integrations connecting** scraping and extraction services into Stacksync's platform.
- **Engineered** multiple additional API integrations on Google Cloud Run, supporting varied authentication methods and schema-driven dynamic inputs.
- **Normalized and validated AI-extracted data** by enforcing schema-driven contracts, improving consistency across downstream integrations.
- **Refactored** backend logic into job ID-based workflows, enabling stateless request chaining and webhook delivery.
- **Collaborated** with GTM and product teams to align API integration workflows with customer use cases, ensuring seamless adoption.

PROJECTS

VoiceCompanionAI >> Backend Engineer

February 2026 – Current [GitHub](#)

Python, FastAPI, Async SQLAlchemy, PostgreSQL, pgvector, Docker, OpenAI APIs

- **Designing a real-time voice** interaction backend integrating wake-word detection (Porcupine), speech-to-text, and LLM-based reasoning into an async job-driven pipeline.
- **Implemented a PostgreSQL-backed job queue** using SELECT FOR UPDATE SKIP LOCKED to enable safe concurrent worker processing without task duplication.
- **Built a long-term memory engine** using pgvector embeddings to store, rank, and retrieve conversational context.
- **Developed intent routing and structured** interaction flows for storytelling and guided conversational activities.

Propia >> Backend Engineer

October 2025 – Current

Python, FastAPI, Async SQLAlchemy, PostgreSQL, Docker, PaddleOCR, AWS S3

- **Designed state-aware lease parsing and chunking profiles** to improve embedding accuracy and semantic retrieval precision.
- **Built an async FastAPI backend with job-based pipelines** (API → queue → GPU OCR worker → database), enabling non-blocking document ingestion.
- **Implemented document-aware embedding pipelines** enabling semantic lease search and structured clause retrieval.
- **Optimized embedding generation and PostgreSQL** storage workflows to improve semantic retrieval accuracy and response latency.

Rabbit Aviation >> Backend Engineering Project (AeroSkyOps)

April 2022 – July 2024 [GitHub](#)

Led backend development of AeroSkyOps, an internal platform for managing aircraft assignments, fuel workflows, and flight tracking.

Flask, SQLite, PostgreSQL, AWS, Docker

- **Replaced an unstructured MongoDB** setup with PostgreSQL, introducing relational integrity for aircraft assignments, reducing manual errors, and improving database query efficiency.
- **Developed RESTful APIs** with Flask and SQLAlchemy to integrate near-live flight tracking and real-time weather updates via the FlightAware API, providing up-to-date aviation insights based on user-input tail numbers.

EDUCATION

APP ACADEMY >> Software Engineering

Completed June 2024

Coding Dojo >> Python, MERN Stack

Completed April 2022